Nishankar Sathiyamohan

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Research Interests

Computer Vision, Trustworthy AI, Spatial Intelligence.

Education

University of Moratuwa — Moratuwa, Sri Lanka
M.Sc. by Research in Computer Science and Engineering — *Sep 2026 – Present*

University of Peradeniya — Kandy, Sri Lanka
B.Sc. (Hons) in Computer Engineering, CGPA: 3.50/4.00 — *Nov 2018 – Dec 2023*

Hartley College — Point Pedro, Sri Lanka
G.C.E. Advanced Level – Physical Sciences — *Jan 2009 – Aug 2017*

- Results: Combined Mathematics (A), Chemistry (A), Physics (B).

Publications

Published Journal Articles

J1. T. Shyamalee*, **S. Nishankar***, S. Thuseethan, C. Wimalasooriya, and Y. Sebastian, "Deep Semi-supervised Learning for Medical Image Analysis: A Survey", *(*Equal Contribution)*, **Neurocomputing** | Q1 | DOI — 2026

J2. Tharindu Godage, **Sathiyamohan Nishankar**, Selvarajah Thuseethan, Shanmuganathan Vasanthapriyan, and Zhongwei Liang, "Trends, Challenges and Future Directions in Deep Learning for Citrus Leaf Disease Recognition: A Survey", **Artificial Intelligence Review**, Springer | Q1 | DOI — Jun 2026

J3. **S. Nishankar**, V. Pavindran, T. Mithuran, S. Nimishan, S. Thuseethan, and Y. Sebastian, "ViT-RoT: Vision Transformer-Based Robust Framework for Tomato Leaf Disease Recognition", **AgriEngineering**, MDPI, vol. 7, no. 6, p. 185 | Q1 | DOI — Jun 2025

J4. **S. Nishankar**, T. Mithuran, S. Nimishan, V. Pavindran, and S. Thuseethan, "TOM-SSL: Tomato Disease Recognition using Pseudo-labelling based Semi-supervised Learning", **AgriEngineering**, MDPI, vol. 7, no. 8, p. 248 | Q1 | DOI — Jul 2025

Manuscripts Under Review or Revision

M1. **S. Nishankar**, P. Sanjeewani, A. Perera, and S. Thuseethan, "HiLRP: Conservation-Valid Attribution via Attention Primitives", **Pattern Recognition**, Elsevier | Under Review | arXiv:2609.01282 — Aug 2026

M2. **S. Nishankar**, N. Pathirana, P. Sanjeewani, A. G. Perera, and S. Thuseethan, "Does Explainability Transfer? A Controlled Benchmark of Attribution Methods on Vision Transformers and CNNs", **Neurocomputing**, Elsevier | Under Review | arXiv:2608.02396 — Jul 2026

M3. **S. Nishankar**, P. Sanjeewani, and A. Perera, "M3-Score: Fidelity, Memorization and Coverage as Separate Axes for Evaluating Generative Radiology Image Models", **Biomedical Signal Processing and Control**, Elsevier | Submitted | arXiv:2609.33769 — Sep 2026

M4. George, R., **S. Nishankar**, Thuseethan, S., Wimalasooriya, C., Sebastian, Y., Ragel, R. G., and Liang, Z., "U-FedTomAtt: Ultra-lightweight Federated Learning with Attention for Tomato Disease Recognition", **Information Sciences** | Under Review | arXiv:2602.16749 — Feb 2026

M5. George, R., **S. Nishankar**, Thuseethan, S., and Ragel, R. G., "UtVAA: Ultra-tiny Vision Transformer with Affix Attention for Mobile Image Classification", **IEEE Transactions on Pattern Analysis and Machine Intelligence** | Under Review | arXiv:2606.14735 — Jun 2026

M6. Thilini Bakmeedeniya, Selvarajah Thuseethan, Roshan G. Ragel, Palanisamy Vigneshwaran, and **Sathiyamohan Nishankar**, "From Unimodal to Multimodal Deep Physiological Signal Analysis in Healthcare: A Survey", **Computer Science Review** | Minor Revision — Mar 2026

M7. Romiyal George, **Sathiyamohan Nishankar**, Selvarajah Thuseethan, Kandiah Pakeerathan, and Roshan G. Ragel, "SLIF-Brinjal: An In-Field Leaf Dataset for Disease Recognition in Precision Agriculture", **Scientific Data**, Nature Portfolio | Minor Revision — Feb 2026

Published Conference Papers

C1. M. Shakishnavi, K. P. N. Jayasena, and **S. Nishankar**, "A Unified Zero Trust Orchestration Framework for Kubernetes: Design, Integration, and Functional Verification", **APAN Conference** | IEEE Xplore — 2026

C2. M. Shakishnavi, K. P. N. Jayasena, and **S. Nishankar**, "A Multi-Layer Zero Trust Security Framework for Kubernetes", **MERCon** | DOI — 2026

Current Drafts

S. Nishankar, et al., **"SSA-HQNN: Spatial Superpixel Attention-Guided Hybrid Quantum Neural Network for Image Classification"** (Drafting). — 2026

- Proposed SSA-HQNN, a hybrid quantum–classical architecture that uses learned spatial superpixel attention to condition trainable quantum encoding for image classification.
- Evaluated across seven benchmarks, achieving up to 93.9% accuracy on PathMNIST and 90.3% on PneumoniaMNIST, while matching or exceeding the AUC of classical baselines.
- Designed a parameter-efficient backbone requiring as few as 60K trainable parameters, substantially fewer than ResNet-18.

S. Nishankar, et al., **"Keeping Minority Opinions in Review Summaries: Do Keyword Prompts Help Small Language Models?"** (Drafting). — 2026

- Investigated whether keyword-guided prompting helps small language models (1B–8B) retain minority opinions in imbalanced review summaries. Tested 12 prompt conditions on five instruction-tuned models, with hypotheses fixed before any run.
- Generated and scored over 27,000 summaries of Amazon and Drugs.com review sets, with splits from 4:4 to 7:1. Opinion-grouped keyphrases increased minority-opinion coverage by 11–38 percentage points over a length-matched baseline across all five models, but pushed summaries toward a false even balance.
- Built a reproducible, resumable evaluation pipeline on a single 8 GB GPU, using NLI-based faithfulness and sentiment-based fairness metrics, Holm-corrected paired tests, entity-clustered bootstrap intervals, and 500+ unit tests.

Honors & Awards

Best Researcher Award (Faculty Level) — Sabaragamuwa University, 2025

1st Place: Pre Aces Hackathon – Inter-University Coding Competition — 2022

Professional Experience

Research Assistant — Sep 2026 – Present
University of Moratuwa, Sri Lanka — *Supervisor: Prof. D. A. Meedeniya*

- Conducting research on occlusion-aware multimodal 3D perception and agentic reasoning for wildfire fuel mapping using 3D point clouds.

Part-time Research Intern Mar 2026 – Aug 2026
University of Southern Queensland, Australia *Supervisor: Dr. Asanka Perera*

- Developed **M3-Score** for evaluating generative medical image models using features from a frozen, radiology-pretrained vision transformer.
- Conducted a controlled benchmark of attribution methods across vision transformers and CNNs.
- Developed **HiLRP**, a conservation-valid attribution framework for vision transformers.

Lecturer Apr 2025 – Aug 2026
Sabaragamuwa University of Sri Lanka *Dept. of Software Engineering*

- Taught undergraduate courses and supervised final-year research projects.
- **University Business Linkage (UBL) Coordinator:** Facilitated collaboration with industry partners.
- **Assistant Research Coordinator:** Coordinated departmental research activities.

Lecturer Apr 2024 – Apr 2025
University of Jaffna *Dept. of Computer Engineering*

Instructor Jan 2024 – Apr 2024
University of Peradeniya *Dept. of Computer Engineering*

Freelance Software Developer 2022 – Present
Remote, Self-Employed

- Developed **DUCAS AI** to automate daily construction reporting using Azure, GPT-4o, and FastAPI.

Software Engineering Intern Dec 2022 – Apr 2023
Infinity Innovators Pvt *Colombo, Sri Lanka*

- Developed **WAKA**, a full-stack online vehicle marketplace for New Zealand, using React.js and AWS.

Teaching Experience

Sabaragamuwa University of Sri Lanka – *Lecturer* 2025 – Aug 2026

- Taught Artificial Intelligence, Computer Communication Networks, High Performance Computing, Cloud Computing, Game Design and Development, and Parallel and Distributed Computing.

University of Jaffna – *Lecturer on Contract* 2024 – 2025

- Taught Embedded Systems, Computing, Digital Image Processing, and Applied Algorithms.

University of Peradeniya – *Instructor* 2024

- Led instruction in Artificial Intelligence, Computer Communication Networks, and Advanced Database Management Systems.

Projects

Efficient Transfer Learning and XAI for Histopathology 2023

- Applied SHAP, LIME, and Grad-CAM to examine predictions of histopathology classification models.

Road Sign Detection in Low-Light Conditions 2023

- Developed a YOLOv8-based system to detect road signs in low-light conditions.

Academic Service & Memberships

Publication Chair	ComURS 2026; ICARC 2026
Open Track Co-Chair	ICARC 2026
Registration Chair	ICARC 2026
Editor	ICARC 2026; ComURS 2026; ICSUSL 2025
Reviewer	ICARC 2026; ComURS 2026; MERCon 2026
Member	Institute of Electrical and Electronics Engineers (IEEE)

Technical Skills

Programming Languages: Python, Java, C++, SQL
Backend & Application Development: Node.js, Express.js, Django, Flask, FastAPI, React.js, Flutter
AI/ML: PyTorch, TensorFlow, Keras, OpenCV, Scikit-learn, YOLOv8, Hugging Face, Nerfstudio
Cloud & DevOps: AWS (Serverless, EC2, S3), Azure, Docker, Git

Referees

Dr. S. Thuseethan

- Lecturer in Information Technology, Charles Darwin University, Australia.
- Email: Thuseethan.selvarajah@cdu.edu.au

Dr. Asanka Perera

- Lecturer in Mechatronic Engineering, University of Southern Queensland, Australia.
- Email: Asanka.Perera@unisq.edu.au

Dr. Pubudu Sanjeewani

- Associate Lecturer, RMIT University, Melbourne, Australia.
- Email: pubudu.sanjeewani@rmit.edu.au

Prof. D. A. Meedeniya

- Professor, Department of Computer Science & Engineering, Faculty of Engineering, University of Moratuwa, Katubadda, Moratuwa, Sri Lanka.
- Email: dulanim@cse.mrt.ac.lk / dulanim@uom.lk